Evaluating moving target defence against advanced persistent threat attackers modelled on documented intrusion campaigns

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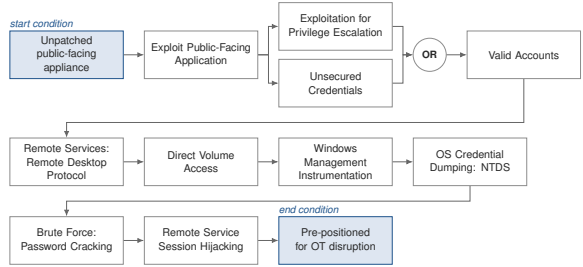
Supervisor: Dr Jin B. Hong

Honours seminar, The University of Western Australia

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MTD is proposed against APT attackers, but rarely tested against one

- **Advanced persistent threat (APT)**: one target, months to years
- Volt Typhoon: five years inside US infrastructure [CISA 2024]
- **Moving target defence (MTD)**: keep the network moving
- Tested against simple, scripted attackers

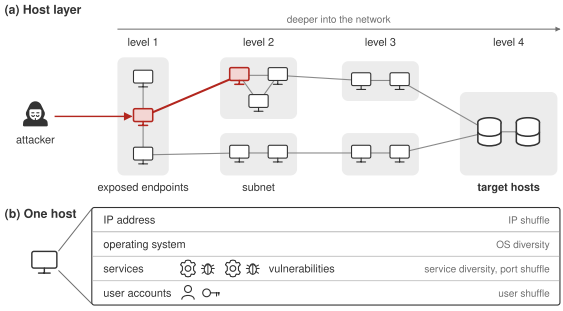


Volt Typhoon's intrusion as an **attack flow**

Research question: How does MTD perform against APT attackers?

MTDSim: seven MTD mechanisms, one scripted attacker

- **MTDSim** [Brown 2023; Zhang 2023; Tay 2024]
- **Host layer**: where hosts are
- **Service layer**: what runs on them
- **Baseline attacker**: six **attack actions**, fixed order



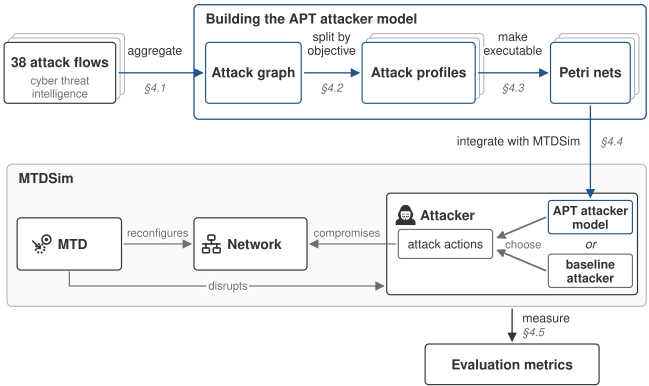
Recent MTD attacker models lack most APT properties

	Property	Baseline attacker 2023	Masud et al. 2025	Kim et al. 2026
1	Campaign persistence	✓	part	part
2	Objective-specific behaviour	✗	✗	✗
3	Choice of attack strategy	part	✗	✗
4	Adaptivity to MTD disruption	part	✗	✗
5	Detection avoidance	✗	✗	part
6	Cost–benefit reasoning	✗	✗	✗
7	Learning from past deployments	✗	✗	✗
8	Deployment-strategy awareness	✗	✗	✗

1, 2, 4, 5: the APT definition [Alshamrani 2019; NIST 2011]. 3, 6, 7, 8: two MTD surveys [Cho 2020; Jalowski 2026].

Gap: how MTD performs against APT attackers is unmeasured.

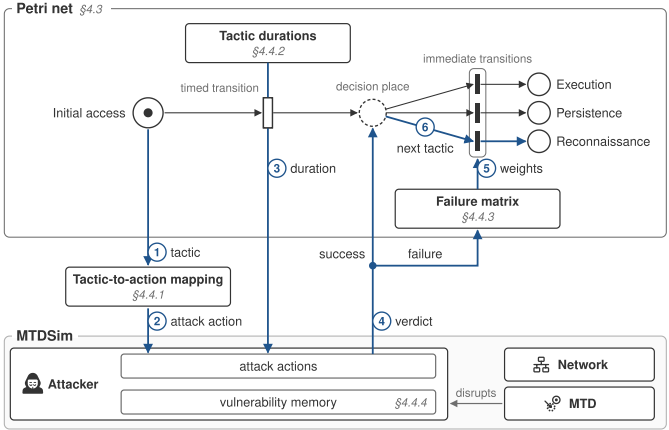
Build an APT attacker model from threat reports; run it in MTDSim



- **SQ1**: 38 attack flows → four **attack profiles** by objective
- **SQ2**: each runs as a Petri net
- **SQ3**: two attackers, same MTD

c1 stole data
c2 impeded systems
c3 both
c4 neither

A Petri net picks the next tactic; MTDSim carries it out



- Next tactic: drawn from the attack flows
- 7 of 15 tactics **unmapped**
- Timing and failures: declared

Experiment: both attackers on the same 1 000 networks

Attacker	baseline; APT model, c1 to c4
MTD	none; 7 mechanisms; 3 deployment strategies
Interval	50 to 2 000 s
Seeds	1 000 per configuration

NCR: share of hosts compromised

NCR reduction

$$= 1 - \frac{\text{NCR under MTD}}{\text{NCR with no MTD}}$$

No MTD: the APT attacker model acts more like an APT, but compromises less

	APT attacker model	Baseline attacker
Distinct openings (first 8 steps)	179 to 947	3
Attack actions per minute	0.68 to 1.23	1.71
Not flagged by a port-scan rule	89 to 98 %	82 %
Runs reaching a target host (ASP)	0.04 to 0.14	0.66
Share of hosts compromised (NCR)	0.15 to 0.21	0.50

Same cause for both halves: its tactic durations and the order of its tactics.

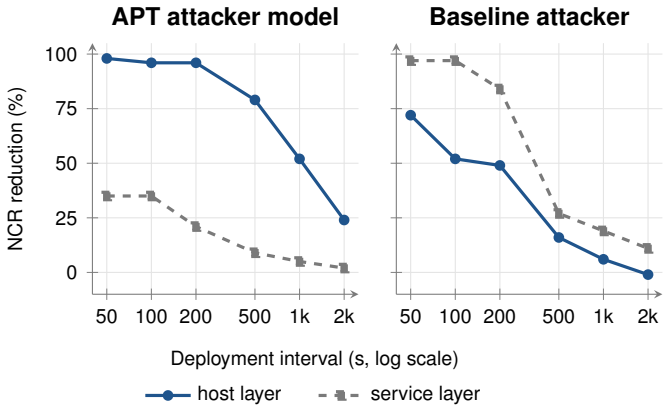
Headline: the most effective MTD mechanism differs between the attackers

NCR reduction at 200 s	APT attacker model	Baseline attacker
host layer		
IP shuffle	**96 % (1)**	68 % (2)
complete topology	**95 % (1)**	42 % (3)
host topology	**95 % (1)**	38 % (3)
service layer		
service diversity	36 % (4)	**92 % (1)**
port shuffle	20 % (6)	**91 % (1)**
OS diversity	7 % (7)	68 % (2)
credentials		
user shuffle	−10 % (8)	−5 % (5)

(rank) among all ten MTD settings

- Rank 1 shares no mechanism
- Service diversity: 92 against 36 %
- Spearman's $\rho = 0.01$

The difference holds at every deployment interval



Why (measured at 2 000 s):

- The service layer disrupts only port scans, exploits and brute force
- Blocks an attack action: 95 % baseline, 15 % APT model
- Host layer costs the APT model 242 to 324 s

The attacker model can decide which MTD looks best

MTDShield, learned against the baseline attacker

NCR reduction, 200 s	Baseline	APT model
MTDShield	**68 %**	28 %
random deployment	33 %	**75 %**

- Test a learned strategy on an attacker it was not trained on
- Report a ranking with its attacker model
- Cost depends on the attacker model

One simplified attacker cannot tell you how MTD performs against an APT.

Limitations

- One simulator, one network size, no deployment cost
- Six attack actions: no data theft, no IDS
- Declared durations and failure matrix
- 38 attack flows
- Cause not isolated

In MTDSim, the most effective MTD depends on the attacker model

- **Built**: an APT attacker model from 38 attack flows, runnable in MTDSim
- **Found**: host layer against it, service layer against the baseline attacker
- **Next**: richer attack actions, an IDS, deployment cost

Evaluate MTD against attacker models grounded in documented behaviour.

Questions?

How does MTD perform against APT attackers?

In MTDSim, it depends on which attacker it is tested against.

Marc Labouchardiere • Supervisor: Dr Jin B. Hong

Code and experiments: github.com/marc-la/MTDSim

Backup: why the APT attacker model compromises less with no MTD

- Unmapped tactics take 34 to 57 % of an attack profile's run, and 15 to 30 % of its attack actions fail because their preconditions are missing (e.g. exploit before scan port)
- Rerun with the baseline attacker's times in place of its tactic durations, same order of tactics: ASP rises from 0.09 to 0.45, against the baseline attacker's 0.66, closing 63 % of the gap
- The order of its tactics most likely accounts for the rest
- The rankings compare each attacker with its own NCR with no MTD, so the lower starting point is unlikely to decide which mechanism ranks first

Dissertation Section 6.1, Appendix G.3, threat T2. Appendix G analyses were chosen after the results were seen (T9).

Backup: footholds under the host layer

- Under the three host-layer mechanisms at 50 to 200 s, 70 to 89 % of the APT attacker model's runs compromise no host, against 3 to 20 % of the baseline attacker's
- With no MTD: 1 % and none
- So the host layer disrupts the APT attacker model mostly *before* its first foothold: after a host-layer deployment it has no host until its Petri net next reaches Lateral Movement
- From 500 s the share falls for both attackers, and the host layer still ranks first against the APT attacker model

Dissertation Appendix G.4 (post hoc, T9); the claim stands on Chapter 5 without it.

Backup: are the declared tactic durations driving the result?

Mean duration (band)	low	declared	high
4.5 s (2.25 to 9)	0.165	0.163	0.159
35 s (17.5 to 70)	0.172	0.163	0.145
36 s (18 to 72)	0.170	0.163	0.148
45 s (11.25 to 180)	**0.267**	0.163	**0.059**

NCR with no MTD, each value moved alone, 400 runs per cell.

- Bands fixed before the run; one value moved at a time
- Only the 45 s (Persistence, Stealth, Command and Control) moves NCR much: Command and Control is the only tactic that finds new hosts
- At a quarter and four times the 45 s, rank 1 still shares no mechanism with the baseline attacker's, and the host layer still ranks above the service layer ($\rho = 0.03$ and -0.03)
- Untested: interactions between values

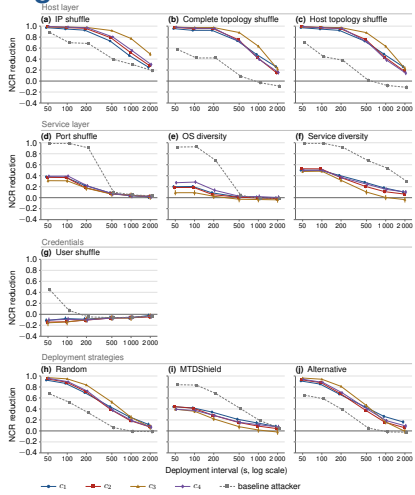
Dissertation Appendix D.1, threat T4.

Backup: what each component of the APT attacker model changes

Component removed	MTD	NCR with	without	Cohen's d
Attack profiles (use the whole attack graph)	no MTD	0.171	0.189	-0.28
	IP shuffle, 200 s	0.007	0.013	-0.47
	OS diversity, 200 s	0.159	0.174	-0.26
Failure matrix	no MTD	0.171	0.167	$+0.07$
	IP shuffle, 200 s	0.007	0.006	$+0.13$
Vulnerability memory	no MTD	0.171	0.169	$+0.04$
	service diversity, 200 s	0.109	0.106	$+0.07$

- Bold: not negligible ($|d| > 0.2$). The failure matrix changes *where* the attacker goes after a failure (back to Reconnaissance 76 to 83 % vs 8 to 10 %), not what it achieves
- Without the attack profiles, the same three host-layer mechanisms still rank first

Backup: each attack profile against the baseline attacker



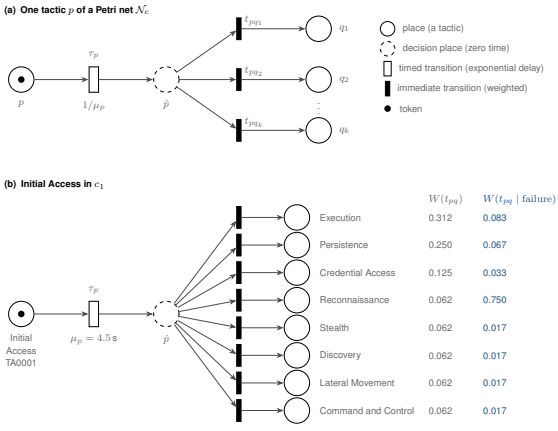
At 50 to 200 s the four attack profiles differ from one another less than the baseline attacker differs from the nearest of them (dissertation Figure 5.4).

Backup: scoring the eight properties

Attacker model	1 Pers.	2 Obj.	3 Strat.	4 Adapt.	5 Detect.	6 Cost	7 Learn.	8 Deploy.
Baseline attacker (2023)	✓	✗	part	part	✗	✗	✗	✗
Masud et al. (2025)	part	✗	✗	✗	✗	✗	✗	✗
Kim et al. (2026)	part	✗	✗	✗	part	✗	✗	✗
APT attacker model	✓	part	✓	part	part	✗	✗	✗

- Each property split into predicates; ✓ = every predicate true of the model's own behaviour; part = a behaviour predicate true, but set by a fixed rule or assumed
- One scorer, checked by blind scoring with language models (dissertation Appendix B)
- Adaptivity is in part because the failure matrix is a fixed rule of mine; detection avoidance in part because MTDSim has no IDS

Backup: the Petri net



Generalised stochastic Petri net [Ajmone Marsan 1995], extended with the failure matrix F:

$$(P, T, I, O, W, M_0, F)$$

Base weight of the move from tactic p to q:

$$W(t_{pq}) = \frac{n(p,q)}{\sum_{q'} n(p,q')}$$

with $n(p,q)$ the attack flows that draw $p \to q$. After a failed attack action:

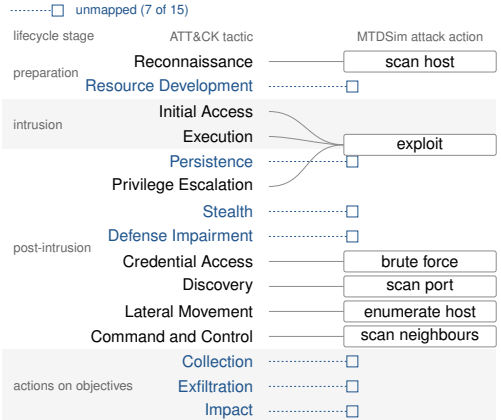
$$W(t_{pq} \mid \text{failure}) = \frac{W(t_{pq})\, F(p,q)}{\sum_{q'} W(t_{pq'})\, F(p,q')}$$

One token, starting on Reconnaissance.

Backup: tactic durations and the tactic-to-action mapping

Tactics	Mean (s)
Initial Access, Privilege Escalation, Credential Access, Lateral Movement (MTDSim's exploit)	4.5
Reconnaissance, Discovery (MTDSim's three scans)	35
Persistence, Stealth, Command and Control	45
Collection, Exfiltration, Impact	36
Execution, Defense Impairment	22.5
Resource Development	0

Last four rows: judgement, kept on the baseline attacker's time scale and under the 200 s default interval.

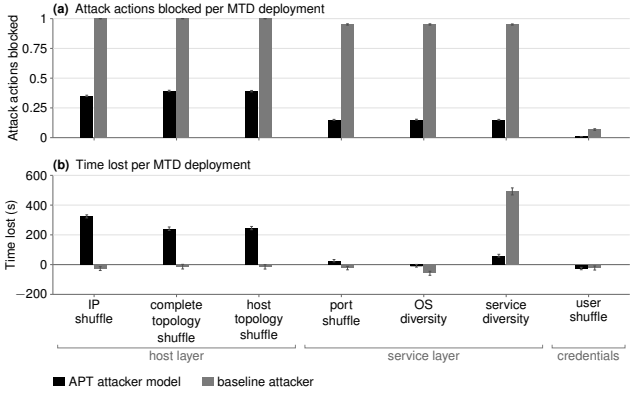


Backup: MTDShield as run

- Run as released by Tay (2024); correcting the release would make it my agent, not Tay's
- Trained at one deployment interval, against the baseline attacker, on 100-host networks
- Known limits of the release: published scores produced with exploration at one; four evaluation inputs read as zero; "do nothing" never a learning target
- Its detection input reads an unmapped tactic as "nothing detected"
- So the result bounds MTDShield *as released*, not MTDShield retrained against the APT attacker model

Dissertation Appendix E, threat T12.

Backup: disruption per MTD deployment, and ASP



- Measured at 2 000 s, because the next deployment caps time lost at the interval
- The baseline attacker's time lost is at or below zero after the first deployment: sent back to scan host, it gets a second attempt at its first host, as MTDSim intends
- On ASP reduction the layer order also holds against each attacker (Table F.3)

Backup: verification, and why no comparison with earlier MTDSim numbers

- An event tracer replays any run step by step, across Petri net, attacker and MTD
- MTDSim's code audited against what its papers describe; unit tests of the Petri net and the simulator
- Fixed seeds reproduce every run exactly
- Defects found in MTDSim were repaired before the evaluation, so results are not compared numerically with Zhang (2023) or Tay (2024)

Dissertation threat T3.

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